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Paleomagnetic results from Triassic sections in the Ordos Basin, North China

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ABSTRACT

Continental Triassic sediments with good fossil control have been sampled in two thick sections (Hancheng, 35.5°N, 110.2°E and Tongchuan, 35.2°N, 109.1°E) from the southern margin of the Ordos basin, Shaanxi Province, China. Detailed analysis of 633 samples coming from 69 sites has been carried out with progressive alternating field and/or thermal demagnetization. Most sites display a low unblocking temperature component up to 200–400°C, which is undistinguishable from the Present Earth Field (PEF) direction, and a characteristic component with higher unblocking temperature with both normal and reversed polarity, which differs from the PEF and later Mesozoic paleomagnetic directions. The pole position from the lower Triassic part of the Hancheng section is at 61.9°N, 348.3°E ($A_{95} = 3.8°$, $N = 16$ sites). The directions of the characteristic magnetization of the middle Triassic sites from the Tongchuan section pass the fold test and are consistent with those from middle Triassic sites of the Hancheng section, giving a mean paleopole at 59.8°N, 359.9°E ($A_{95} = 4.0°$, $N = 17$ sites). The late Triassic sites from the Tongchuan section yield a pole at 62.3°N, 7.7°E ($A_{95} = 3.8°$, $N = 11$ sites). The early Triassic pole from the Ordos basin is consistent with poles obtained from Shanxi province. A late Permian to late Triassic apparent polar wander path of the North China block (NCB) is proposed. It is found that poles from the NCB and from the two adjacent blocks (South China block and Siberia) were widely separated until at least the end of the Triassic, pointing to major motion between the blocks probably during the Jurassic. At the end of the Triassic, the two Chinese blocks were already at similar latitudes but still were to undergo a $78 \pm 28°$ relative rotation. Both were some $38 \pm 4°$ to the south of Siberia (calculated for a reference point at 45°N, 115°E on the NCB/Siberia boundary).

1. Introduction

China and South-East Asia are divided into a mosaic of blocks, some of the most prominent being North China (NCB), South China (SCB), Tarim and Indochina, which became accreted to the Siberian craton from late Paleozoic until possibly mid Mesozoic time (Fig. 1) [1–8]. These blocks are separated by deformed zones including fold belts and fault zones, some with ophiolites and other indicators of former sutures. The detailed history of accretion of the different continental blocks is still strongly debated. For instance, based on geological observations, the collision between North China and Siberia is believed to have occurred in the uppermost Permian according to Burrett [1], but near the Jurassic–Cretaceous boundary according to Klimetz [3]. The age of suturing of the NCB and SCB is not well established either. It occurred during the Devonian according to Mattauer et al. [5], the late Triassic according to Wang et al. [6], Sengör et al. [7] and Hsü et al. [8], and would not have been completed until the early Cretaceous according to Klimetz [3].

Paleomagnetism can provide independent constraints on the history of these blocks. Recent paleomagnetic surveys [9–16] indicate that important relative motions between the NCB and SCB appear to have taken place after the Permian, and that both blocks were located far to the south of their present positions relative to Siberia during Permo-Triassic time. Suturing would have been achieved in the middle Jurassic [11–17] with, how-

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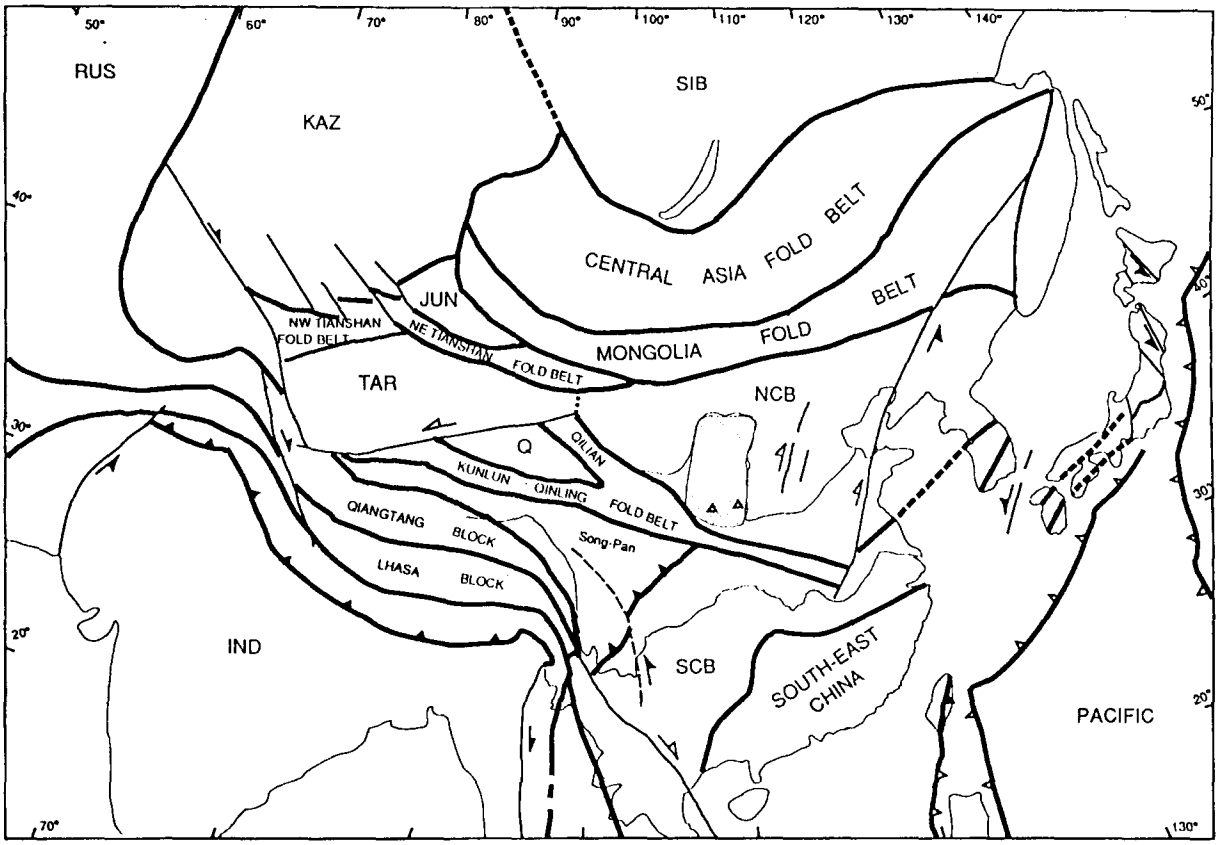


Fig. 1. Sketch map of tectonic units of China. The major blocks are: *RUS* = Russia, *KAZ* = Kazakhstan, *SIB* = Siberia, *TAR* = Tarim, *NCB* = North China block, *SCB* = South China block, *Q* = Qaidam, *JUN* = Junggar, and *IND* = India. The sections sampled are indicated with triangle symbols. The Ordos basin is shaded.

ever, possible subsequent shortening between NCB, SCB and Siberia induced by the India/Asia collision [18–20]. Until recently, few reliable data of middle Triassic to Jurassic age were available for either the NCB or SCB that could further constrain the timing of the final collisions. Moreover, serious problems of Cretaceous overprinting have been pointed out by Kent et al. [21] in the eastern part of China. For these reasons, in an attempt to obtain data for these key periods, we have sampled Triassic continental sediments along two sections from the Ordos basin, in the western part of North China (Figs. 1 and 2).

2. Geological setting and sampling

Late Paleozoic and Mesozoic outcrops are widely distributed in the western part of North China, at the periphery of the Ordos basin (Fig. 2). They consist of well-developed, mainly clastic continental sediments, ranging in age from the Permian to the early Cretaceous. The series is gently folded and allows extensive stratigraphic study and good paleontological control, based on flora and continental vertebrates [22–24]. We have extensively sampled the Triassic sediments along two sections near Hancheng (35.5°N, 110.2°E) and Tongchuan (35.2°N, 109.1°E), on the southeastern margin of the basin (Figs. 1, 2).

2.1. The Hancheng section (lower to middle Triassic)

Lower Triassic sediments rest concordantly over late Permian sediments, with a slight dip (10–20°) towards the west and northwest. They are divided

into two formations, the Lioujiagou and Heshanggou formations [22,23]. The Lioujiagou Formation mainly consists of massive purplish coarse-grained sandstones interbedded with conglomerates, purple mudstones and argillaceous sandstones. They contain plant assemblage, including *Pleuromeria jiaochengensis* Wang, *P. rossica* Neubeurg, *Cramatopteris* sp., and *Yuccites* sp. [22]. The Heshanggou Formation lies conformably over the Lioujiagou Formation, and consists of red to purple mudstones and grey sandy mudstones interbedded with purple sandstones and sandy conglomerates. Vertebrate fossil assemblages including *Fugusuchus hejiapanensis*, *Ceratodus heshanggouensis*, Benthosuchidae and Capitosauridae indicate an early Triassic age.

The middle Triassic sediments of the Zhifang Formation are continuous with the lower Triassic series, and consist mostly of massive purple to red and purple-grey sandy mudstones, with vertebrate fossils and plants, such as *Sinokannemeyeria*, *Parakananemeyeria*, *Shansisuchus*, *Pseudosuchia* sp., *Darwinula gerdae* and *Spirorbis* cf. *aberrans*.

2.2. *The Tongchuan section (middle to upper Triassic)*

The middle Triassic sediments of the Tongchuan section have a similar lithology to those of the Hancheng section, consisting of purple to red argillaceous sandstones with good fossil control. The series is gently folded, and was sampled on both flanks of an anticline, with a N110°E trending axis. The north flank has a shallow dip of 7°, while the south flank shows steeper values around 45°. Folding is thought to be early Jurassic to early Cretaceous: indeed, two unconformities can be distinguished, the first one between late Triassic and early Jurassic sediments, and the second one between middle Jurassic and early Cretaceous sediments. The gently folded overlying upper Triassic sediments are subdivided into the Tongchuan and Yanchang Formations [23]. The Tongchuan Formation consists of a sequence of green-grey massive feldspathic sandstones, interbedded with some black shale levels. These sediments contain plant fossils such as *Tongchuanophyllum concinum*

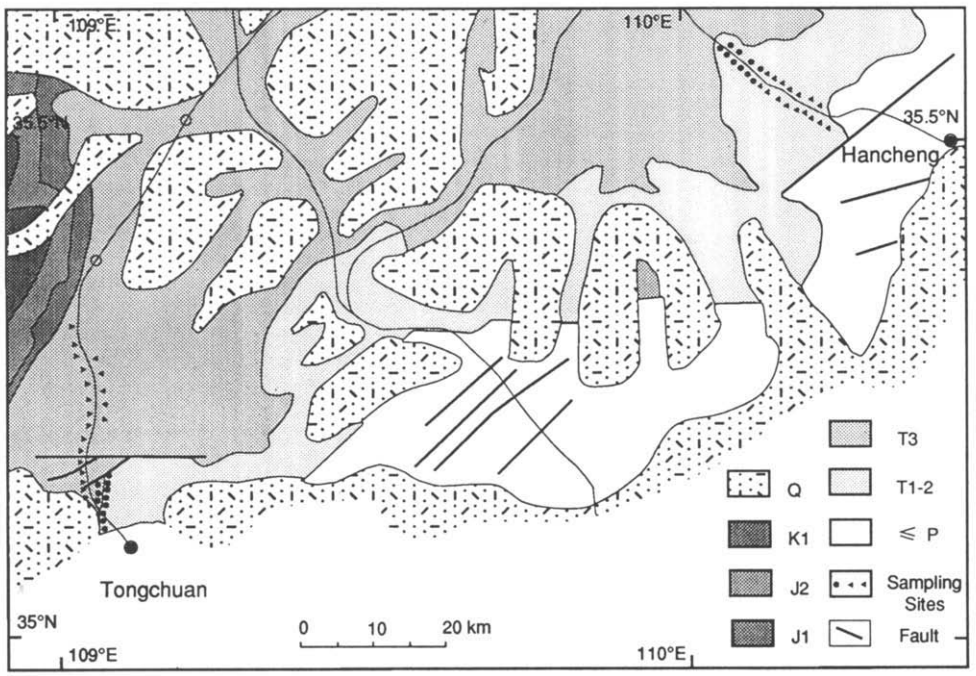


Fig. 2. Simplified geological map of sampling area. Sites along the two sampled sections are shown as open triangles (lower Triassic), filled circles (middle Triassic) and filled triangles (upper Triassic). *P* = Permian, *T1-2* = early/middle Triassic, *T3* = late Triassic, *J1* = early Jurassic, *J2* = middle Jurassic, *K1* = early Cretaceous, and *Q* = Quaternary.

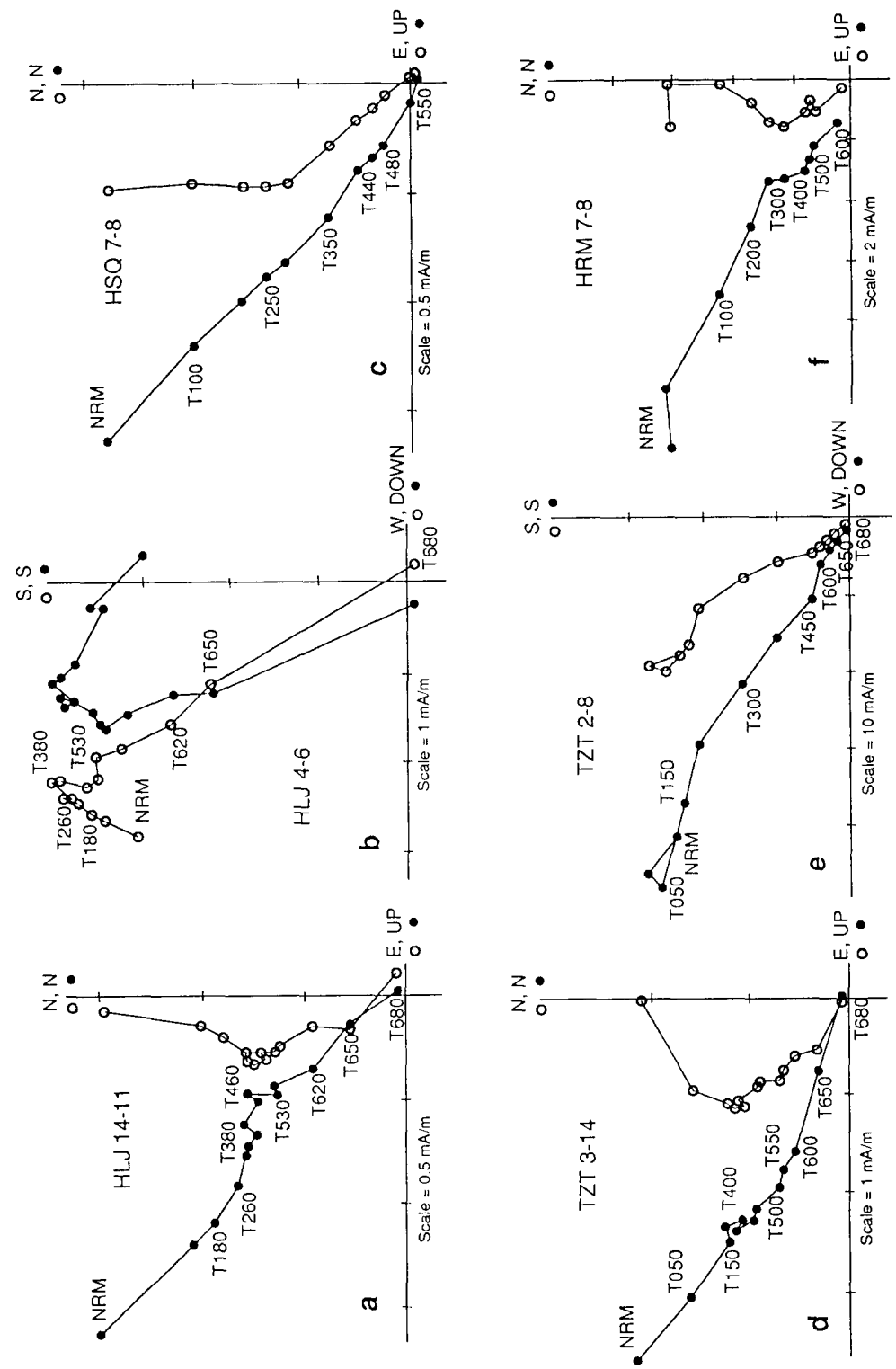


Fig. 3. Orthogonal projections of representative progressive thermal demagnetizations for early (a, b and c) and middle (d, e and f) Triassic samples. All plots are in geographic coordinates. The open (resp. filled) symbols refer to the horizontal (resp. N–S vertical) plane. Note that the coordinate system is flipped for samples with reverse polarity characteristic directions.

(Zhou and Huang), *Danaeopsis magnifolia* (Zhou and Huang), *Pleuromeria tongchuanesis* (Zhou and Huang), *Glossopteris chinesis* (Zhou and Huang), *Neocalamites* sp., *Equisetites*, and *Sporopollen* assemblages including *Punctatisporites*, *Granulatisporites gigantus* and *Chordasporites*. The Yanchang Formation is essentially a coal measure with plant assemblages such as *Daneopsis fecunda* (Halle), *Bernoullia zeilleri* (Pan), *Todites Shensiensis* (Pan), *Glossophyllum? shensiensis* Sze. The upper Triassic sediments are mainly fine greyish to green feldspathic sandstones and sandy shales with interbedded coal series.

2.3. Sampling

The samples from the two sections were collected using a portable, gasoline-powered drill, and oriented in situ using a magnetic compass, or a sun compass when possible. Each site consists of five to twelve individually oriented cores which were generally distributed over three to ten meters of the outcrop. Thirty-seven sites were collected from the Hancheng section (178 oriented samples, from 27 sites of early Triassic age and 10 sites of middle Triassic age) and 32 sites from the Tongchuan section (334 samples from 12 sites of middle Triassic age and 20 sites of late Triassic age). The sampled cores, 2.5 cm in diameter, were cut in the laboratory into 2.2 cm long specimens for paleomagnetic measurements.

3. Laboratory work and results

Specimens were separated into two collections and were measured separately at the paleomagnetic laboratory of the Institut de Physique du Globe de Paris (IPGP) and the Institute of Geomechanics in Beijing, using respectively a three-axis cryogenic magnetometer (CTF) and a Schonstedt DSM-2 spinner magnetometer. In both cases, some samples were demagnetized by alternating fields using a Schonstedt GSD-1 demagnetizer, most, however, being demagnetized thermally. At least ten heating steps were carried out (up to 680°C), followed by cooling in field-free space (less than 10 nT). At the

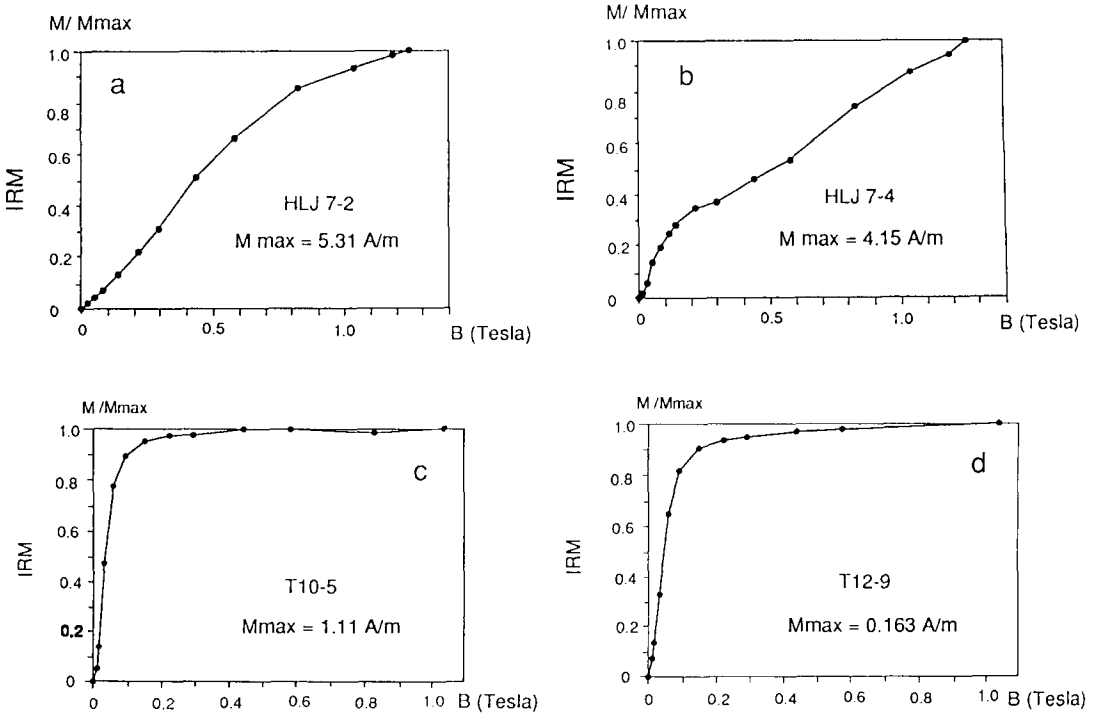


Fig. 4. Typical isothermal remanence magnetization (IRM) acquisition curves of red (a, b) and grey (c, d) specimens from lower and upper Triassic specimens respectively. (a), (c), and (d) are prior to heating; (b) after heating to 680°C.

Paris laboratory, the bulk magnetic susceptibility of the specimens was measured after each heating step in order to monitor possible magnetic mineral transformations.

Magnetic directions were plotted on Zijderveld vector diagrams [25] to identify discrete magnetic components which were analyzed using principal component analysis [26]. Site mean directions were calculated using Fisher statistics [27] when directions only were obtained within a site, or by McFadden and McElhinny's method when individual characteristic directions had to be combined with great circle intersections with sector constraints [28].

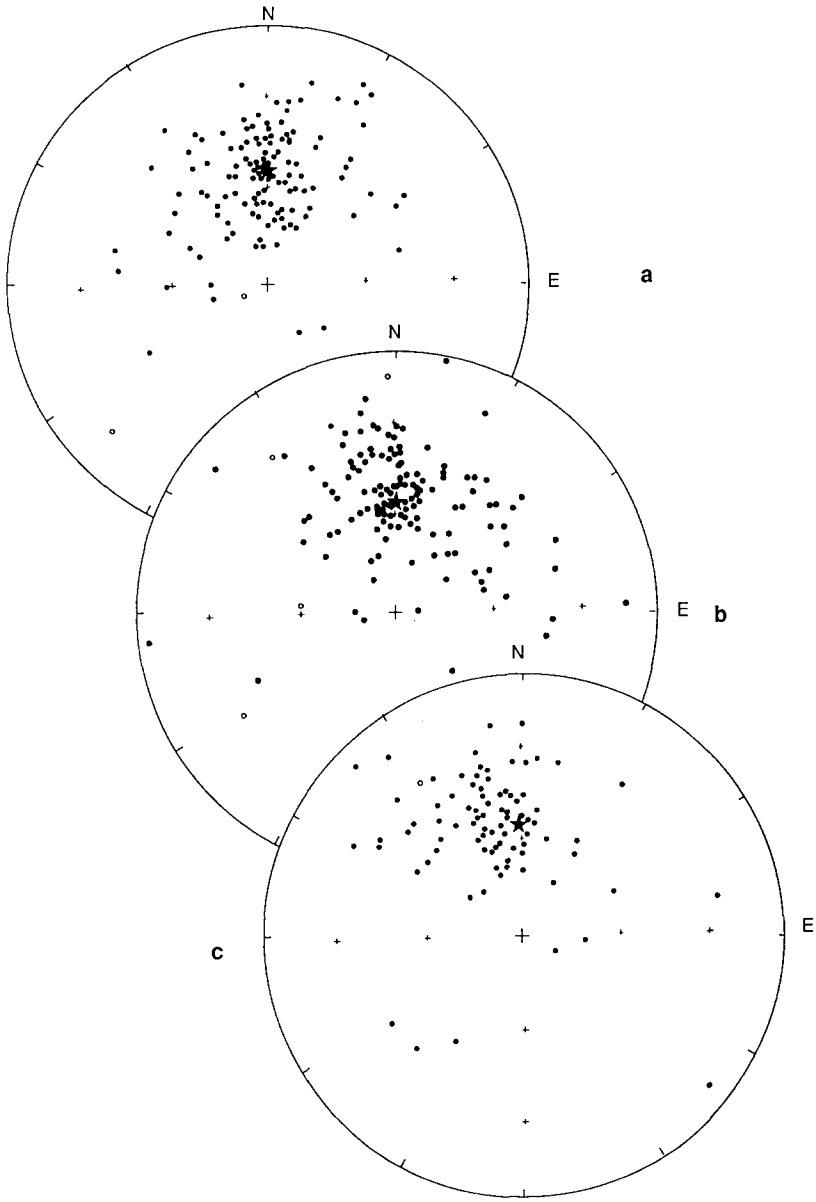


Fig. 5. Equal-area stereographic projection of low-temperature components from specimens of (a) early, (b) middle and (c) late Triassic age in geographic coordinates. The filled (resp. open) circles refer to the lower (resp. upper) hemisphere. The star represents the recent geomagnetic dipole field direction.

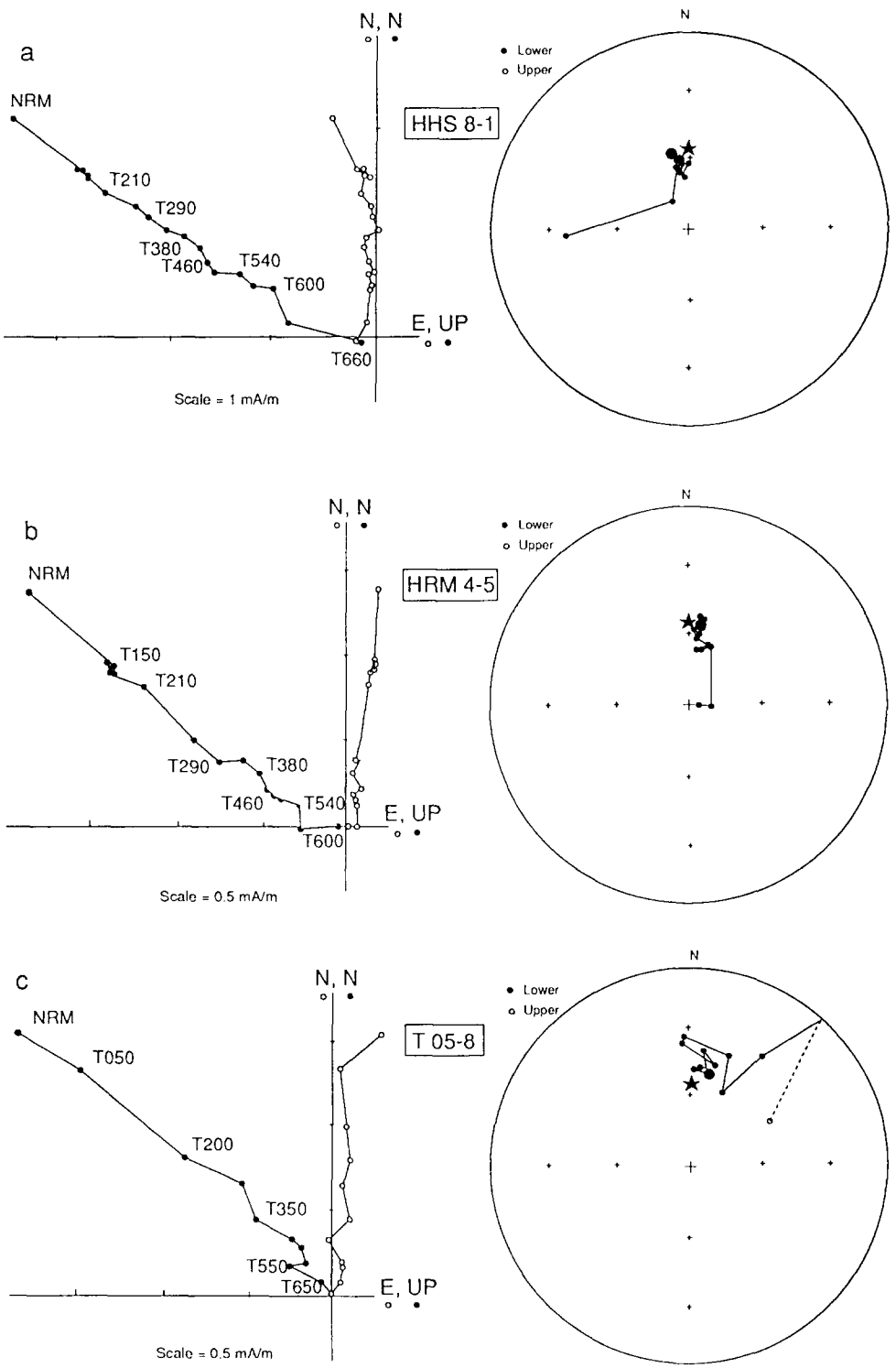


Fig. 6. Demagnetization diagrams for grey sandstones (orthogonal and stereographic projections in geographic coordinates) showing a strong recent geomagnetic field overprint, with no possibility to isolate any high-temperature component. (a), (b) and (c) are from early, middle and late Triassic specimens respectively.

3.1. *Rock magnetism*

Two main categories of rocks were measured: red and grey sandstones.

Most red bed specimens (85%) have NRM intensities ranging from 1 to 5 mA/m, with NRM directions generally clustering around the present earth field (PEF) direction. Alternating field demagnetization was insufficient and thus most samples were thermally demagnetized. A first component, unblocked between 180° and 400°C, was easily removed by thermal demagnetization (Fig. 3a, b). It is seen on isothermal remanent magnetization (IRM) acquisition curves (Fig. 4a, b) that saturation is not reached after applying a field of 1.4 T for the red sandstones, indicating, along with the 680°C maximum unblocking temperatures (Fig. 3a, b), that hematite is the main magnetic carrier. Figure 4b shows a small component saturated around 0.2 T suggesting that magnetite has been formed after heating to 680°C.

The grey sandstones have a weaker intensity (65% of the samples have an NRM under 1 mA/m) but with exceptions going up to 40 mA/m. They also have a low temperature overprint, visible for instance in Fig. 3c, f between NRM and 300°C. This component is in the direction of the PEF in geographic coordinates (Fig. 5). Some grey-sandstone specimens have a strong overprint in the direction of the PEF (Fig. 6). About 50% of these specimens display this PEF overprint up to an unblocking temperature of 300°C, beyond which the scatter of directions is such that high-temperature components cannot be determined. These specimens were discarded before further analysis. IRM acquisition curves from unheated grey specimens (Fig. 4c, d) reveal the existence of a low-coercivity mineral. Saturation is reached after applying a field of 0.2 T, and these specimens are demagnetized at 550–600°C, indicating that magnetite is probably the main carrier of the remanence.

Variations of magnetic mineral species during heating were monitored by susceptibility measurement after each step. Most of the red sediment specimens show a relatively stable susceptibility

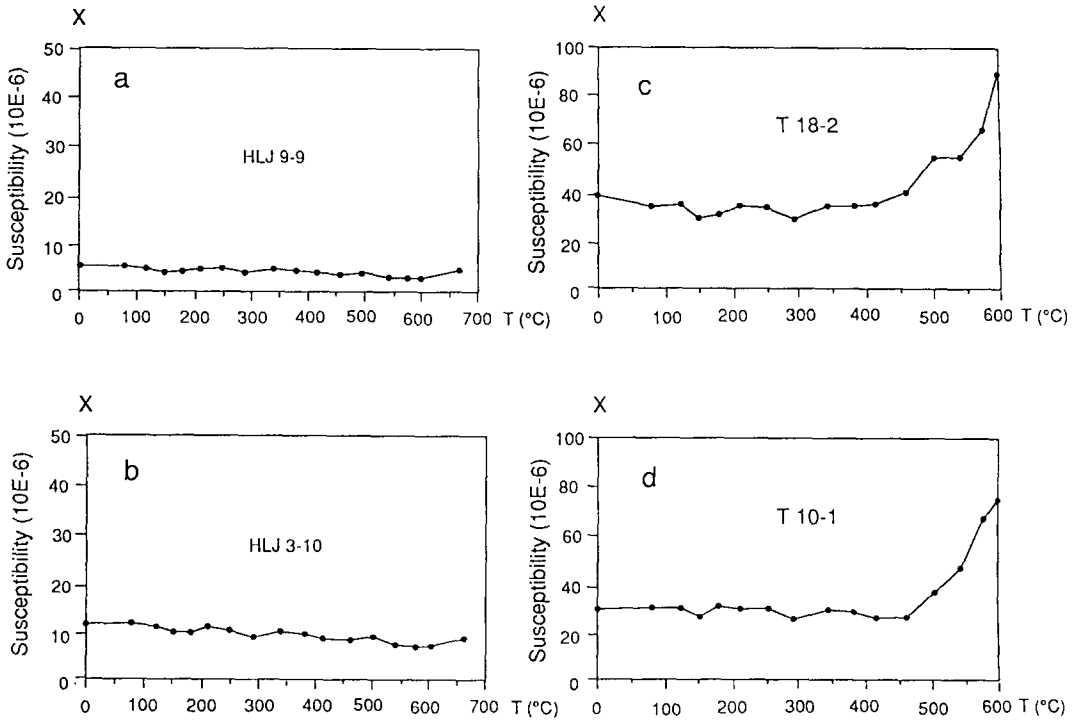


Fig. 7. Evolution of susceptibility with thermal treatment. Susceptibility remains roughly constant for specimens of lower Triassic red sandstones from the Hancheng section (a, b), showing little mineralogical change during heating. Susceptibility increases after 450°C in some specimens of upper Triassic grey sandstones from the Tongchuan section (c, d), indicative of mineralogical changes (see text).

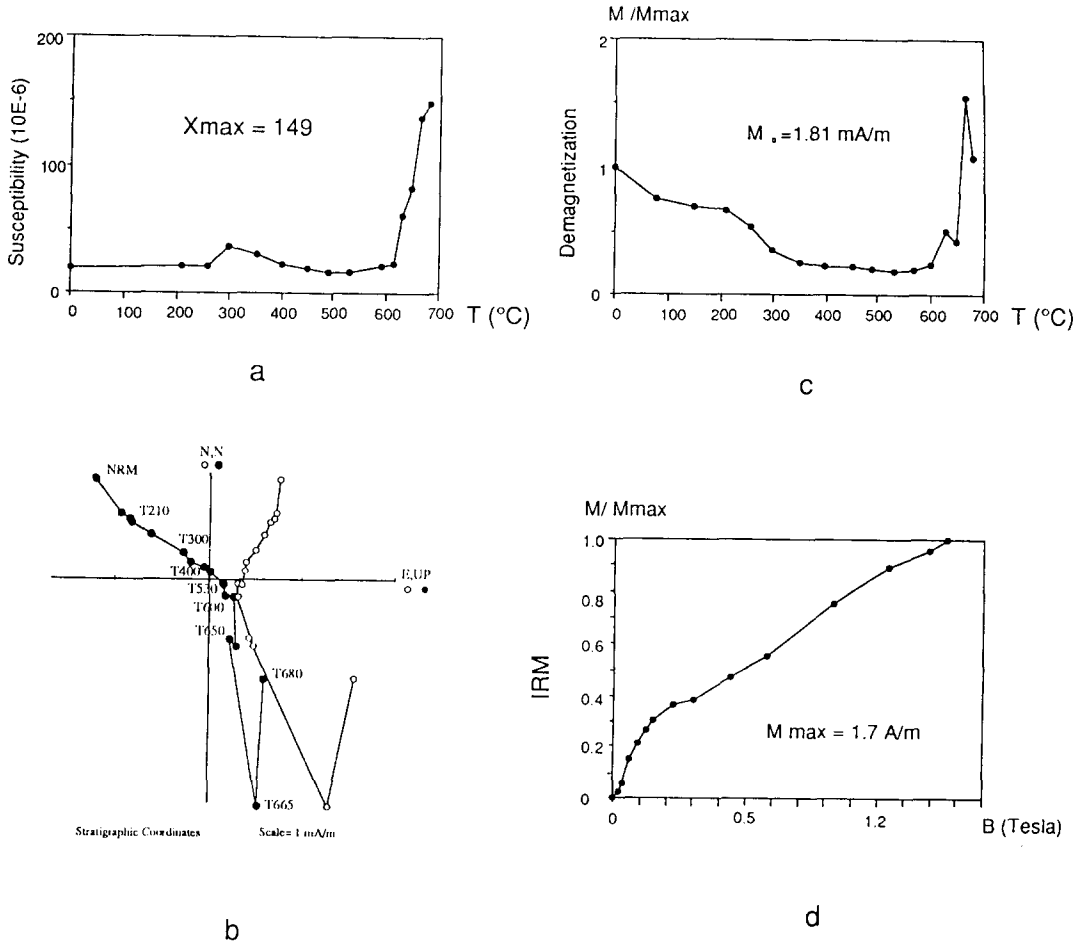


Fig. 8. One example of changes in magnetic mineralogy of a middle Triassic specimen from the Hancheng section. Bulk susceptibility (a) and remanence (b) change drastically after heating at about 600°C, i.e. close to the unblocking temperature of magnetite. Vector diagrams (c) show a consistent direction until that temperature. Mineral changes are obvious beyond and SIRM experiments of the heated (680°C) sample (d) clearly indicate both magnetite and hematite.

(Fig. 7a, b), where the susceptibility of grey sandstones may increase close to the Curie temperature of magnetite (Fig. 7c, d). Some grey sandstones show a major change in susceptibility at high temperature (about 600°C and above, e.g. Fig. 8a; specimen from the Hancheng section), and the IRM acquisition curves of heated samples show that hematite has been produced at high temperature, although the dominant magnetic mineral still is magnetite (Fig. 8b, d).

3.2. *Lower Triassic*

The lower-temperature component from the lower Triassic is close to the PEF direction (local dipole field direction $D = 0°$, $I = 56°$), with a mean direction of $D = 359.4°$, $I = 55.9°$ ($\alpha_{95} = 4.7°$, $n = 130$ samples) in geographic coordinates, corresponding to a recent overprint (Fig. 5a). A high-temperature component is often clearly visible after 500°C and not destroyed until 650–680°C in red samples (Fig. 3a, b); in grey samples, the range is from 400° to 550–580°C (Fig. 3c). In both rock types, both polarities are observed and demagnetization generally follows a path towards to origin.

The mean direction from the 16 early Triassic sites of the Hancheng section was usually computed using Fisher statistics [27], magnetic directions of individual specimens being determined by

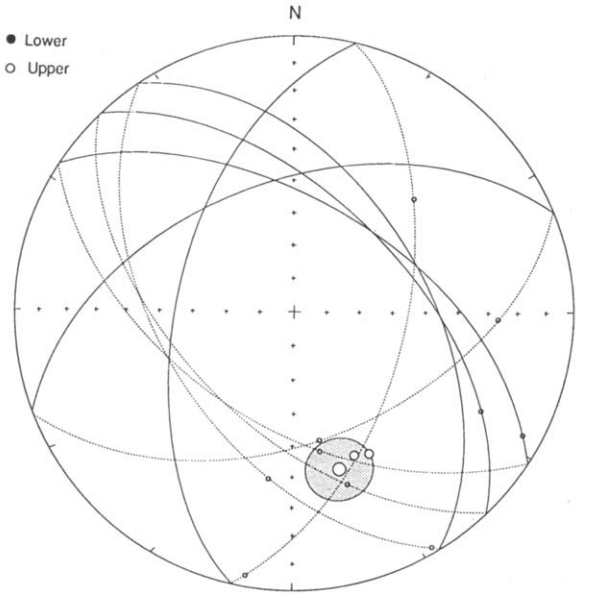


Fig. 9. An example of combined remagnetization great circles and directions of 7 samples from lower Triassic site HLJ 3. Great circles are shown with limiting sector contraints [28]. Triangles correspond to directions determined with principal component analysis [26]. The overall mean direction is shown with its circle of 95% confidence (darker shading). Stereographic projection with open symbols in the upper hemisphere.

principal component analysis. About 15% of these directions become erratic and it is difficult to define linear segments with principal component analysis either due to a weak magnetization intensity close to the unblocking temperature or to mineralogical changes at high temperature. Remagnetization circles can then often be determined and combined with direct determinations of remanent directions, using McFadden and McElhinny's method [28] to calculate site-mean direction (Fig. 9). The mean direction of the high-temperature components (HTC) for the early Triassic sites are shown in Fig. 10a. They are directed to the NNW (down) and SSE (up) and form two antipodal clusters. The section has a shallow monoclinal dip, so no fold test is possible. The overall mean direction is $D = 337.1°$, $I = 33.5°$ ($\alpha_{95} = 4.9°$, $N = 16$) before and $D = 335.2°$, $I = 32.5°$ ($\alpha_{95} = 4.2°$, $N = 16$) after tectonic correction (Table 1). The decrease in α_{95} is not statistically significant at the 95% level [29].

3.3. *Middle Triassic*

Samples from 12 sites of the Tongchuan section, and 10 sites from the Hancheng section have a middle Triassic age; they also show a lower-temperature PEF secondary component ($D = 3.5°$, $I = 55.5°$, $\alpha_{95} = 5.2°$, $n = 137$ samples) (Fig. 5b). High-temperature components could be isolated in only 6 sites of the Hancheng section and 11 sites of the Tongchuan section, with dual polarities (Fig. 3d, e, f) and mean directions different

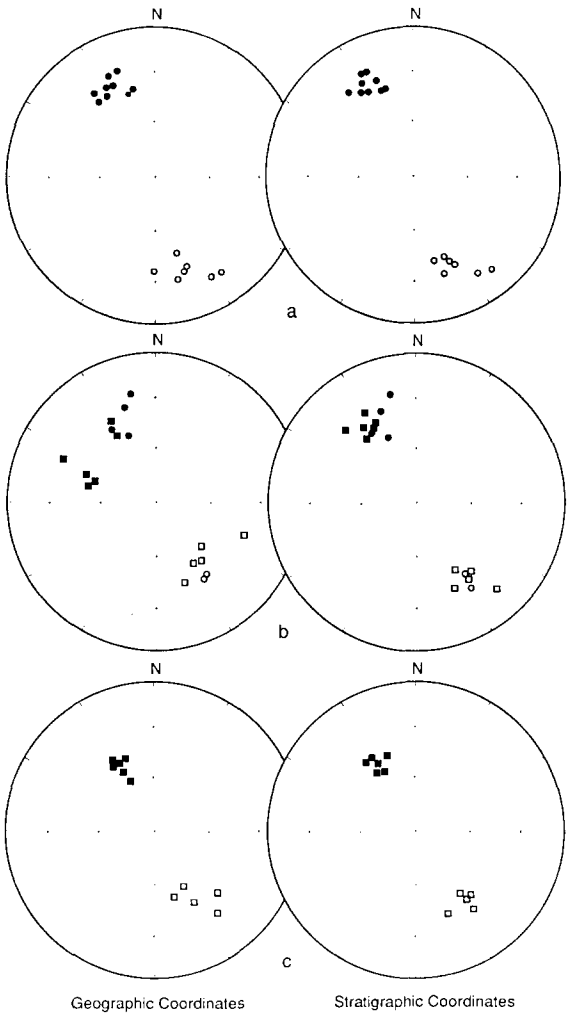


Fig. 10. Equal-area stereographic projection of site-mean characteristic directions, plotted prior to (left) and after (right) tectonic correction: (a) early Triassic, Hancheng section (dots); (b) middle Triassic, Hancheng (dots) and Tongchuan (squares) sections; (c) late Triassic, Tongchuan section (squares). Filled and open symbols refer to the lower and upper hemispheres respectively.

TABLE 1

Characteristic directions and corresponding virtual geomagnetic pole positions for early Triassic sites of the Hancheng section (35.5°N, 110.2°E)

Site	N (n, c)	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k_s	k_s/k_g	$(\alpha_{95})_s$ (°)	λ (°N)	φ (°E)
HLJ15	7 (7, 0)	324.2	32.1	322.3	30.2	13.4		17.1	51.3	360.2
HLJ14	5 (5, 0)	329.8	37.3	328.7	35.0	24.2		15.9	58.0	358.1
HLJ13	4 (4, 0)	335.5	26.4	335.4	25.5	56.3		12.4	58.7	341.5
HLJ12	5 (5, 0)	343.5	41.6	342.5	39.8	118.6		7.1	70.1	344.7
HLJ11	5 (5, 0)	346.7	39.4	342.0	39.1	12.6		22.5	69.4	344.6
HLJ9 + 10 [a]	8 (2, 6)	161.6	−33.6	157.0	−33.8	15.8		15.5	63.5	356.4
HLJ8	7 (3, 4)	160.0	−36.1	158.5	−36.1	23.0		13.7	65.6	346.6
HLJ7	7 (7, 0)	180.4	−35.6	167.6	−39.7	104.5		5.9	73.1	333.3
HLJ6	5 (5, 0)	145.0	−21.3	141.8	−18.8	20.1		18.0	46.7	353.0
HLJ5	5 (5, 0)	336.3	33.0	339.1	33.0	11.1		25.6	64.5	342.3
HLJ4	5 (5, 0)	149.0	−22.7	147.9	−21.4	26.2		15.2	52.0	348.2
HLJ3	7 (2, 5)	163.3	−45.5	161.2	−40.0	50.9		9.4	69.3	347.3
HLJ2	5 (3, 2)	166.1	−29.3	164.5	−30.0	26.5		17.2	66.2	329.7
HLJ1	8 (8, 0)	332.7	33.2	331.8	30.9	19.0		13.1	58.6	350.6
HSQ8	5 (3, 2)	323.4	37.6	332.6	37.0	22.7		17.5	61.8	355.8
HSQ6 + 7 [a]	5 (4, 1)	340.8	26.0	334.7	25.6	21.2		17.5	58.3	342.5
Average of:										
normal sites	9	334.7	34.3	334.2	30.0	114.8	1.28	4.8	61.2	349.5
reversed sites	7	160.2	−32.5	156.4	−31.7	48.8	1.25	8.7	62.1	345.2
all sites	16	337.1	33.5	335.2	32.5	76.7	1.32	4.2	61.7	347.6
Site VGPs								3.8	61.9	348.3

N: number of samples (n = number of direct observations; c = number of remagnetization great circles); D_g, I_g and D_s, I_s: declination and inclination in geographic and stratigraphic coordinates, respectively; k: precision parameter after tilt correction; α_{95}: half-angle of cone of 95% confidence about the mean direction in stratigraphic coordinate. λ (°N) and φ(°E): latitude and longitude of virtual geomagnetic pole.

[a] Average from two nearby sites.

from the PEF (Fig. 10b). The site-mean directions from the Tongchuan section show more scatter before than after tilt correction. Comparison of the precision parameters (k) before and after tilt correction shows a significant increase, with a positive fold test (Table 2), according to the tests of McElhinny [29] (99% confidence level) and McFadden and Jones [30], or to the new Direction-Correction test of Enkin [31]. The tests confirm that the two limbs shared a common mean direction before folding (Table 2).

The site-mean characteristic directions of the middle Triassic sites after tilt correction from the Tongchuan and Hancheng sections are not significantly different at the 95% level, using McFadden and Jones's [30] method (Table 2). A global middle Triassic average has therefore been determined, yielding a mean direction of $D = 329.5°$, $I = 37.5°$ ($\alpha_{95} = 4.4°$, $N = 17$) (Table 2).

3.4. *Upper Triassic*

Samples of grey sandstones and siltstones of late Triassic age come from the Tongchuan section. Four typical NRM demagnetizations are shown in Fig. 11. About 90% of these grey sandstones with weak magnetization are severely overprinted by the PEF (Fig. 6c). Beyond 300°C heating, their magnetization becomes viscous or very weak; often their directions are unstable and ill-determined and their susceptibility changes (Fig. 7c, d). The mean direction of the LTCs before tilt correction falls close to the direction of the PEF with $D = 347.4°$, $I = 52.5°$ ($\alpha_{95} = 5.6°$, $n = 86$) (Fig. 5c), although it is significantly rotated counterclockwise, in the direction of the higher-temperature component, probably due to incomplete separation of the two (i.e. overlap of unblocking temperature spectra), or lower Jurassic remagneti-

TABLE 2

Characteristic directions and corresponding virtual geomagnetic pole positions for middle Triassic sites

Site	N (n, c)	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k_s	$(\alpha_{95})_s$ (°)	k_s/k_g	λ (°N)	ϕ (°E)
(a) Hancheng section (35.5°N, 110.2°E)										
HRM1	4 (0, 4)	145.6	−40.0	147.0	−32.2	292.5	10.3		55.7	357.2
HRM2	3 (3, 0)	346.9	26.9	347.4	25.1	31.9	22.2		65.0	320.3
HRM3	4 (1, 3)	147.0	−39.0	146.1	−40.8	96.6	12.8		58.3	367.0
HRM6	5 (4, 1)	337.3	50.7	338.0	49.8	27.6	15.3		71.0	11.6
HRM7	6 (6, 0)	327.7	43.5	328.7	43.1	20.5	15.2		61.2	367.7
HRM10	6 (5, 1)	340.8	33.6	340.6	33.7	26.5	13.5		65.8	340.5
Average	6	334.5	39.2	334.9	37.7	51.5	9.4	0.9	63.8	353.7
(b) Tongchuan section (35.2°N, 109.1°E)										
TZT0	3 (3, 0)	111.9	−36.1	137.4	−21.0	22.1	26.9		44.4	357.5
TZT1	7 (7, 0)	294.9	31.1	317.2	31.9	21.8	13.2		48.1	365.2
TZT2	6 (6, 0)	290.5	48.9	330.9	42.1	21.8	14.7		62.3	364.0
TZT3	9 (8, 1)	285.7	53.3	326.1	37.6	24.7	10.6		57.1	362.7
TZT4	7 (7, 0)	284.2	52.4	323.8	44.6	20.9	13.5		57.8	373.1
TZT5	6 (3, 3)	134.7	−54.7	146.4	−37.9	101.2	7.2		57.5	362.8
TZT6	8 (3, 5)	142.7	−48.4	149.6	−45.5	96.4	6.0		62.8	370.0
TZT7	10 (3, 7)	160.1	−41.8	156.0	−37.0	21.3	10.6		64.4	350.8
TZT9	7 (7, 0)	330.5	38.2	331.2	30.3	19.5	14.0		58.1	350.2
T21	8 (8, 0)	329.4	46.8	334.0	39.3	16.5	14.1		63.9	356.4
T22	10 (0, 10)	147.6	−49.5	141.5	−40.4	28.7	10.2		54.5	369.7
Average	11	310.4	47.4	326.5	37.2	86.4	4.9	3.53 *	57.3	361.9
(c) Combined results										
Average of:										
normal sites	10	319.7	45.8	333.1	38.0	73.3	5.7	4.07 *		
reversed sites	7	141.3	−45.2	146.1	−36.6	76.3	7.0	1.96		
all sites	17	319.8	45.1	329.5	37.5	67.0	4.4	2.96 *	59.7	360.2
Site VGPs	17					81.1	4.0	4.74 *	59.8	359.9

Fold test

1. McElhinny's [29] fold test: * indicates positive fold test at the 99% confidence level
2. McFadden and Jones's [30] fold test:
   - $N = 11$ two flanks — Before tilt correction: $P = 10.96$, after tilt correction: $P = 0.945$; F2,18 = 6.013 for 99% level
   - $N = 17$ two flanks — Before tilt correction: $P = 6.378$, after tilt correction: $P = 1.831$; F2,32 = 5.336 for 99% level
   - $N = 17$ three flanks — Before tilt correction: $P = 11.167$, after tilt correction: $P = 1.282$; F4,28 = 4.074 for 99% level
3. DC test [31]:
   - $N = 11$ two flanks — DC slope = 0.936 ± 0.275; positive fold test at 95% level

zation. The characteristic directions from only 11 out of 20 sites could be isolated by principal component analysis or great circle constraints [28]. The distribution of characteristic site mean directions are antipodal and similar to those found in the previous sections (Fig. 10c). Because of gentle folding, the fold test is not positive at the 95% level according either to the McElhinny test [29], or the DC test [31] (Table 3), although there is an indication of improved grouping. The average site mean ChRM direction is $D = 331.5°$, $I = 47.2°$ ($\alpha_{95} = 5.4°$, $N = 11$) before, and $D = 329.6°$, $I = 43.7°$ ($\alpha_{95} = 3.7°$) after tilt correction.

4. Discussion

We have isolated characteristic remanent directions of lower, middle and upper Triassic age from 2 sections in the Ordos basin. Despite the fact that the standard fold test is not possible due to the gentle tilt dip in the Hancheng area, the early Triassic data obtained in this study show a preci-

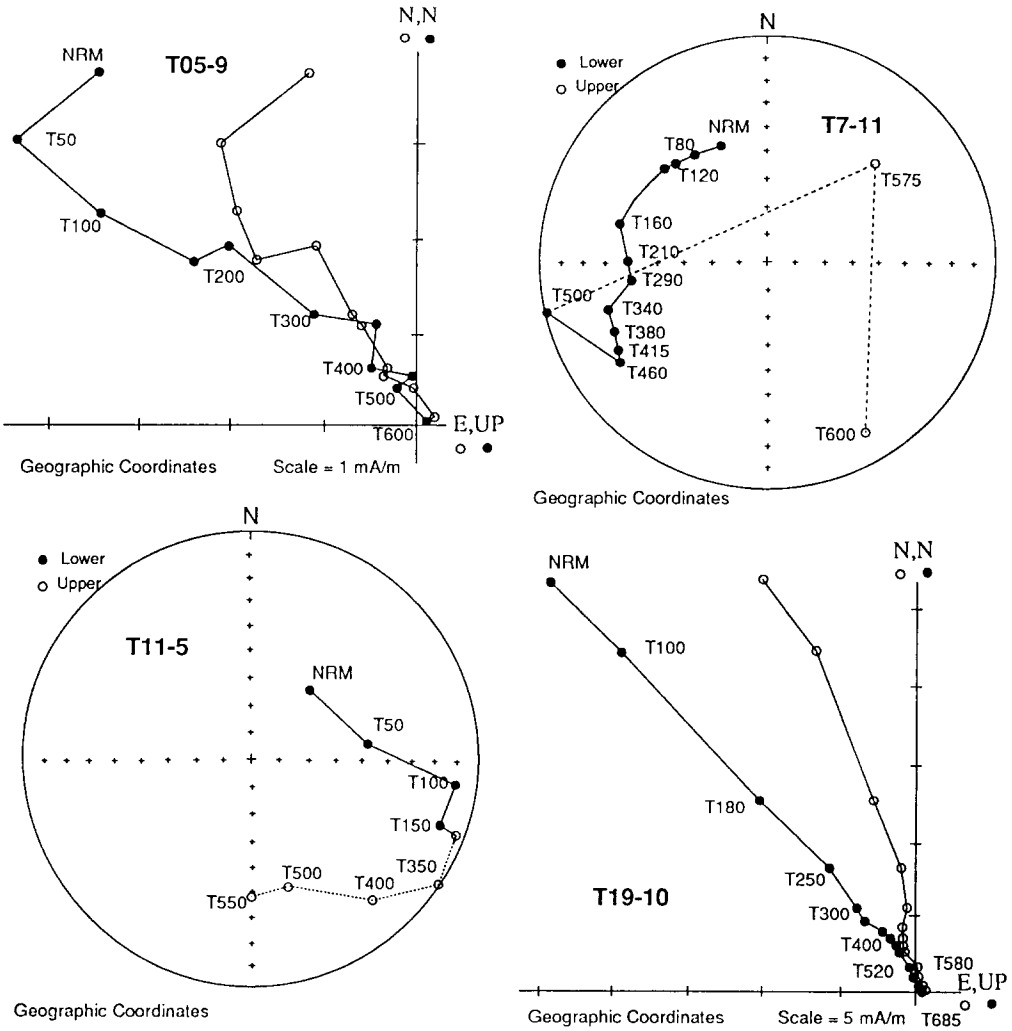


Fig. 11. Representative Zijderveld diagrams and stereographic projections of remagnetization great circles for late Triassic specimens from the Tongchuan section.

sion parameter k increasing with the unfolding process, and pass the reversal test. The early Triassic pole position is at Lat. = 61.9°N, Long. = 348.3°E (A_{95} = 3.8°). The middle Triassic ChRM directions pass the reversal test and the fold test at the 99% confidence level, either using results from the Tongchuan section alone or with the Hancheng section included (Table 2), which suggests that the magnetization is pre-Jurassic to early Cretaceous and probably primary. The pole position calculated from the mean directions of the two sections is at Lat. = 59.8°N, Long. = 359.9°E (A_{95} = 4.0°). The late Triassic ChRM directions are just short of passing the fold test and do pass the reversal test, and differ from both the Jurassic–Cretaceous [19] and recent geomagnetic directions, giving a mean pole position at Lat. = 62.3°N, Long. = 7.7°E (A_{95} = 3.8°). The similarity with the middle Triassic pole, featuring both a reversal and a fold test, gives further confidence in the primary nature of the late Triassic magnetization.

Since the 1981 study of McElhinny et al. [9], a number of paleomagnetic poles of the North China block have been published, covering the upper Permian and lower Triassic [10,11,13,15–17,32]. As a whole, seven upper Permian poles from widely distributed sites (Table 4a, Fig. 12a) yield a mean pole at 48.2°N, 354.1°E (A_{95} = 7.4°). Although

the poles are reasonably consistent, there may be some evidence of streaking along a small circle, possibly indicative of minor local rotations (Fig. 12a).

As far as the lower Triassic is concerned, three relatively reliable poles were available prior to our study (Table 4a, Fig. 12b). The pole from Lin [11] is clearly an outlier, and has a large uncertainty. This may be due to local rotations due to proximity of the Tancheng–Lujiang fault system (Fig. 12b). The two other poles from Shanxi [10] and Shaanxi [13] are consistent with ours, although based on fewer data than the present study. Yet, giving equal weight to these three poles yields a mean lower Triassic pole at 57.7°N, 352.8°E (A_{95} = 6.3°).

Preliminary middle and upper Triassic poles have been reported in the last two years from Shanxi [33] and the Tongchuan area of Shaanxi [34,35]. These results cannot be further discussed for lack of details on sampling sites, demagnetization and treatment methods. The present middle and upper Triassic results, which both pass fold and reversal tests, supersede them.

The NCB appears to have suffered little motion relative to the pole between the upper Permian and the lower Triassic (630 ± 540 km of northward motion and 8 ± 11° of counterclockwise rotation, hardly different from zero, in part due to rather large uncertainties). There is no rotation from the lower to the upper Triassic (0.4 ± 9.2°) and an indication for some northward motion (1000 ± 420 km).

Paleomagnetic data from the same time window for the South China block have been reviewed by Enkin et al. (in [36]) and are given in Table 4b. The upper Permian pole is an average of seven rather consistent studies (A_{95} = 6.2°) of sites distributed over the Emeishan basalts. There are ten studies that are acceptable for the lower Triassic;

TABLE 3

Characteristic directions and corresponding virtual geomagnetic pole positions for late Triassic sites of the Tongchuan section (35.2°N, 109.2°E)

Site	N (n, c)	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k_s	k_s/k_g	$(\alpha_{95})_s$ (°)	λ (°N)	φ (°E)
T2	5 (3, 2)	135.2	−39.4	145.3	−41.4	16.4		20.8	57.9	7.9
T3 [a]	3 (2, 1)	321.7	30.5	330.3	31.5	50.4		29.1		
T4	3 (3, 0)	336.6	59.1	331.8	51.8	305.6		7.1	66.5	20.6
T5 [a]	3 (1, 2)	328.2	34.3	321.3	47.8	13.4		35.2		
T6	4 (2, 2)	143.4	−30.1	141.9	−41.5	206.6		7.4	55.2	10.7
T7 [a]	3 (1, 2)	144.7	−33.6	139.7	−39.7	43.8		31.3		
T8	4 (2, 2)	152.4	−55.1	146.4	−46.3	31.6		19.2	60.7	13.6
T9+10 [b]	6 (4, 2)	151.5	−43.4	145.4	−34.7	14.2		19.0	55.5	0.6
T11	4 (3, 1)	331.6	44.1	326.7	41.3	47.4		14.3	59.0	6.6
T13	5 (4, 1)	334.2	46.4	331.1	46.0	49.3		11.3	64.2	9.6
T14+16 [b]	8 (0, 8)	163.5	−51.1	161.0	−39.3	60.9		8.4	69.0	346.3
T17+18 [b]	6 (5, 1)	336.2	53.0	327.1	50.8	29.9		12.7	62.4	21.0
T19	8 (8, 0)	338.9	47.9	339.2	44.9	113.1		5.2	70.2	358.9
T20	3 (2, 1)	330.1	46.5	330.1	41.5	106.7		14.1	61.8	3.7
Average of:										
normal sites	6	334.4	49.5	331.0	46.1	218.7		4.5	64.2	9.9
reversed sites	5	148.3	−44.2	148.0	−40.8	131.4		6.7	58.8	4.8
all sites	11	331.5	47.2	329.6	43.7	149.8	2.09 *	3.7	62.2	7.3
Site VGPs	11					148.4		3.8	62.3	7.7

Fold test

1. McElhinny's [29] fold test: * indicates positive fold test at the 90% confidence level
2. DC test [31]: the test also indicates a positive fold test (DC slope = 0.945 ± 0.496) at the 95% level

[a] These sites are excluded from further analysis due to $\alpha_{95} \geq 28°$.

[b] Average from two nearby sites.

based mostly on samples from the Sichuan basin, these are consistent and yield a well-defined (A_{95} = 4.7°) mean pole [36]. In contrast, there are very few data for the middle and upper Triassic, and Enkin et al. determined a very uncertain pole from 4 studies (A_{95} = 19.3°) (Table 4).

The NCB and SCB poles are shown in Fig. 13, together with the poles of corresponding age from the Siberian plate [37] (Table 4c). Figure 13 fully vindicates the proposal of Opdyke et al. [12] and Zhao and Coe [15,16] that the three blocks were far from their present configuration in the latest Paleozoic. This conclusion is clearly applicable to the entire Triassic. The relative motions of these blocks during the period from the upper Permian to the upper Triassic were much smaller than their subsequent motions which brought them to their present configuration. In particular, the angular displacement needed to reconcile the NCB and SCB poles in the Permian [15,16] becomes even larger by the end of the Triassic due to anticlockwise rotation of the SCB at the Permo-Triassic boundary [36]. The NCB and SCB poles therefore imply that despite consistent paleolatitudes during this entire period, the two blocks must have undergone a relative rotation of 78 ± 28° after the Triassic. At that time, however, the Chinese blocks were still significantly southward of Siberia (Figs. 13, 14). Figure 14 shows one of many possible reconstructions in the lower Triassic, given the uncertainties in the longitudes. Because of the consistent paleolatitudes and of faunal and floral similarities from Carboniferous to Permian [38], initial collision between the NCB and SCB may

TABLE 4

Permian and Triassic paleomagnetic poles from the NCB, SCB and Siberia block

Site location (°N/°E)	Age	N (n)	Pole position	A_{95} (°) dp/dm	Reference
(a) North China block					
37.8/112.3	Pu	12	44.3°N/357.9°E	3.8/7.3	[9]
39.4/110.0	Pu	(9)	48.6°N/357.0°E	4.8/8.7	[13]
37.5/114.4	Pu	(12)	55.2°N/1.4°E	6.3	[16]
38.6/112.1	Pu	(33)	37.9°N/2.9°E	6.9	[16]
40.1/113.2	Pu	(19)	55.2°N/335.7°E	4.7	[16]
39.6/98.0	Pu	43	41°N/358.6°E	4.3	[32]
43.0/118.9	Pu	11	339.5°N/52.5°E	9.0	[17]
Mean		7	48.2°N/354.1°E	7.4	
37.8/112.3	Tl	(56)	55.5°N/354.1°E	3/5.4	[10]
39.1/110.0	Tl	(15)	55.7°N/355.3°E	5.8/9.9	[13]
41.3/123.7	Tl [a]	6	35.3°N/39.7°E	9.9/15.8	[11]
35.5/110.2	Tl	16	61.9°N/348.3°E	3.8	this study
Mean		3	57.7°N/352.8°E	6.3	
35.2–35.5/109.1–110.2	Tm	17	59.8°N/359.9°E	4.0	this study
35.2/109.2	Tu	11	62.3°N/7.7°E	3.8	this study
(b) South China block					
26.1–29.6/101.9–103.4	Pu [b]	7	53.4°N/244.0°E	6.2	[36]
28.6–32.4/103.4–119.8	Tl [b]	10	45.6°N/218.2°E	4.7	[36]
26.0–32.0/106.0–119.0	Tm + Tu [b]	4	48.7°N/198.2°E	19.3	[36]
(c) Siberian plate					
50–70/61–120	Pu	–	45.0°N/141.0°E	10	[37]
	Tl	–	49.0°N/149.0°E	4	[37]
	Tm-u	–	59.0°N/137.0°E	8	[37]

Age of rock unit is indicated by Pu for late Permian, Tl for early Triassic, Tm for middle Triassic, Tu for late Triassic; N and n are number of sites and samples respectively.

[a] Poles were excluded from average due to probable local rotation (also see text).

[b] Results are from N different studied regions [36].

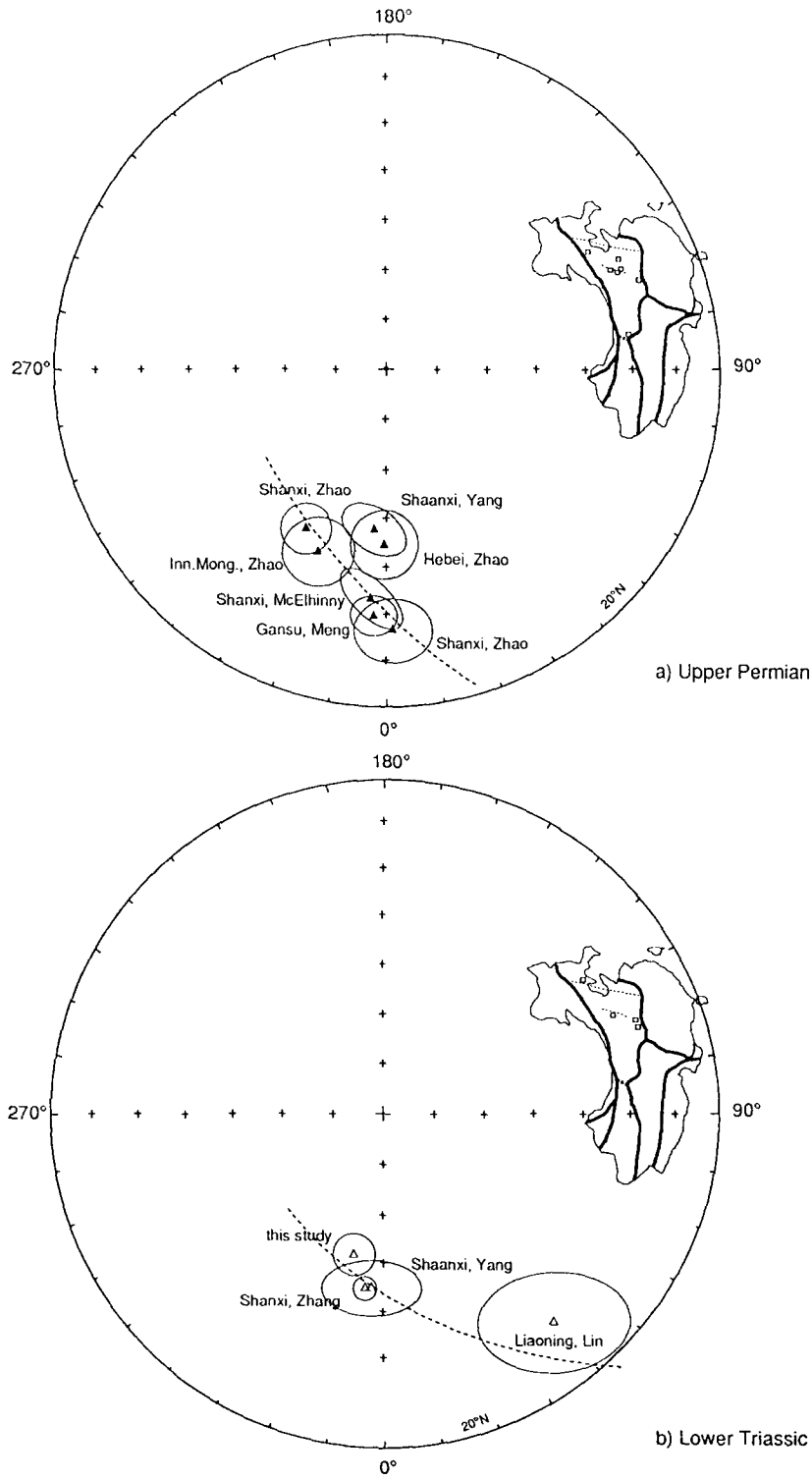


Fig. 12. Paleomagnetic poles and sampling areas of the NCB (simplified tectonic map) from different studies are shown in stereographic projection. (a) upper Permian, (b) lower Triassic.

well have occurred in the late Paleozoic. However, the large angular discrepancies between the poles from the two large blocks imply major tectonic events somewhere along the Qinling and the disappearance of rather large amounts of (oceanic?) lithosphere after the Triassic. There are unfortunately very few Jurassic poles from the NCB and SCB to test and refine a relative motion model. Late Mesozoic poles from the NCB and SCB are consistent [11,20], indicating that the blocks had come to occupy their present relative positions and that the collisional events leading among other things to the formation of the Qinling range were over (although the range was later reactivated by major strike-slip motion due to the India–Asia collision [39]). The large motion hypothesized by Zhao and Coe [15,16] must therefore have occurred essentially during the upper Triassic to Jurassic, which appears contrary to some geologic evidence [5]. As they point out [16], the scissor-like motion is only one of an infinity of possible models. Zhao and Coe have mentioned some geological evidence suggesting active westward-propagating tectonics in the Qinling during the Mesozoic. We can also mention that early to middle Triassic strata in Fung county, Shaanxi Province, are discordantly overlain by late Triassic and Jurassic rocks. Late Mesozoic "detachments" [5,40] and volcanic and magmatic activity of the same age is also reported in the Qinling. Choice

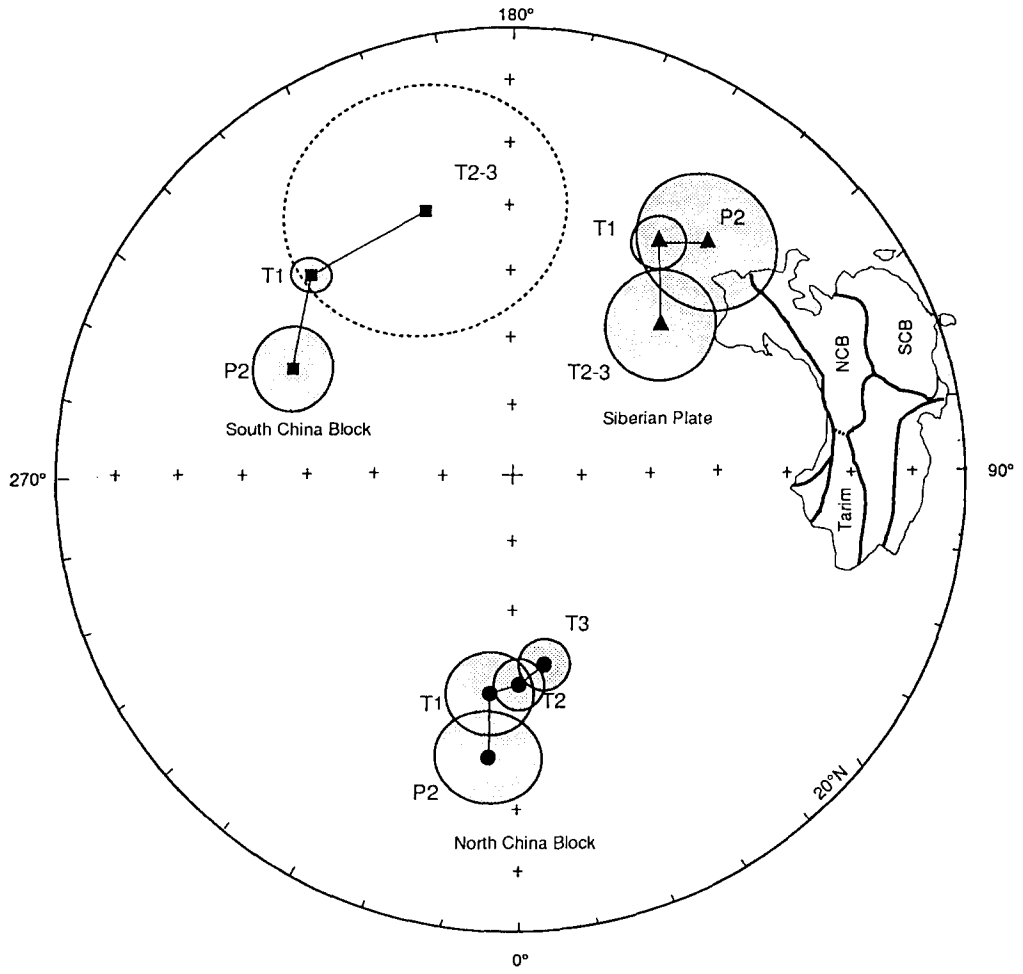


Fig. 13. Stereographic North Polar projection of Permian and Triassic paleomagnetic poles from North and South China [36] and Siberia [37]. *P2* for upper Permian, *T1* for lower Triassic, *T2* for middle Triassic and *T3* for upper Triassic. See text and Table 4. Lower, middle and late Triassic poles for the North China block, indicated by circles are from this study (see also Table 4).

between various models requires better geological constraints and is made easier by working backwards from the Present and invoking continuity arguments. The reader is referred to the synthesis by Enkin et al. (in [36]). Still, it remains somewhat puzzling that the major paleomagnetic discrepancy which persists until the late Triassic was not unambiguously revealed by direct field observation.

Motion between the Chinese blocks and Siberia must also have taken place well into the Mesozoic, this time implying not only relative rotation but also major north–south convergence. This is further discussed by Zhao et al. [17] and Enkin [36].

5. Conclusion

Separate lower, middle and upper Triassic poles are obtained for the Ordos basin. Together with upper Permian and other lower Triassic poles, the new data allow us to outline the APWP of the NCB from the upper Permian to the upper Triassic. Overall motion during this period is rather moderate, with almost no significant rotation and slight northward motion. In contrast, the SCB appears to undergo a rapid anticlockwise rotation during that time. In any case, the poles from NCB and the two blocks presently bounding it to the south (SCB) and north (Siberia) are still very widely separated by the end of the Triassic (angular distances of $68.7 \pm 19.7°$ for NCB/SCB and $56.6 \pm 8.9°$ for NCB/Siberia). Since their Cretaceous poles are more or less concordant [20], this indicates that most of the relative motion must have occurred during the Jurassic, a key period of probable tectonic activity with unfortunately rather scarce data, either paleomagnetic or structural. As was the case when displaced terranes were discovered, paleomagnetism appears to have revealed major relative motions that had gone un-



Fig. 14. A possible paleogeographic reconstruction of the North and South China blocks and their surroundings in the lower Triassic. Relative longitudes of the blocks are unconstrained.

noticed by the field geologists. The NCB and SCB appear to have been at similar paleolatitudes throughout the Jurassic, indicating that continental collision may well have started in the late Paleozoic although it was not completed until the Jurassic. There is an urgent need for Jurassic data from all these blocks. Once again it appears that the ages of a collision may appear distinctly different when viewed from a paleontological, tectonic or paleomagnetic standpoint (e.g. [41,42]). The major post-Permian rotation of the NCB relative to the SCB, first proposed by Opdyke et al. [12] and confirmed by Zhao and Coe [15,16], must have taken place even later than originally proposed by these authors (i.e. our data place stronger constraints on the timing of the motion). The Chinese blocks appear to have been accreted prior to their collision with Siberia, since a significant paleolatitude difference of $38 \pm 4.5°$ (calculated for a reference point at 45°N, 115°E on the NCB/Siberia boundary) still remained in the late Triassic. The subsequent counterclockwise rotation of these Chinese blocks with respect to Siberia is discussed by Zhao et al. [17]. The data presented in this paper are included in a compilation of Chinese paleomagnetic poles from the Permian to the Present, which is accompanied by a series of paleogeographic reconstructions [36].

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I.P.G.P. Contribution No. 1163.

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